

Deep Learning Image Classification Using MNIST

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AI 100
Midterm Project

1. Problem Definition

The goal of this project is to build a deep learning model that can recognize handwritten digits from images. This is a classification problem because the model must classify each image into one of ten categories (digits 0–9).

The MNIST handwritten digit dataset was used for this project.

2. Dataset

The dataset contains:

- 70,000 grayscale images
- 28x28 pixel images
- Digits from 0–9
- 60,000 training images
- 10,000 testing images

The data was normalized by dividing pixel values by 255.

3. Deep Learning Model

A Sequential Neural Network was created using TensorFlow and Keras.

The architecture includes:

1. Flatten layer
2. Dense layer with 128 neurons
3. Dropout layer
4. Output layer with softmax activation

Optimizer:

- Adam

Loss Function:

- `sparse_categorical_crossentropy`

Metric:

- `accuracy`

4. Results

The model achieved approximately 97–98% accuracy on the testing dataset.

The model successfully classified most handwritten digits correctly.

Some mistakes occurred when digits were unclear or poorly written.

5. Lessons Learned

This project helped demonstrate:

- Data preprocessing
- Neural network training
- Model evaluation
- Image classification
- TensorFlow/Keras implementation

It also showed how deep learning can solve real-world classification problems.

6. Conclusion

This project successfully built a deep learning classifier using the MNIST dataset.

Future improvements could include:

- CNN models
- More training epochs
- Additional datasets